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Education

Bachelor of Science in Materials and Metallurgical Engineering

Feb 2020 – March 2025

Bangladesh University of Engineering and Technology (BUET), Dhaka, Bangladesh

Relevant Courses: Crystallography, Phase Diagrams, Electrical & Magnetic Properties of Materials, Mechanical Behavior of Materials, Glass and Ceramic, Materials Processing for Micro & Nano-Systems, Surface Engineering, Computer Programming, Computer Applications to Materials Engineering

GPA: 3.18/4.00

Thesis Title: Effect of Cryogenic Deformation on Properties of AA6063 Alloys

Supervisor: Dr. A. K. M. Bazlur Rashid, Professor, Department of Materials and Metallurgical Engineering, BUET, Dhaka, Bangladesh.

Co-supervisor: Dr. Saquib Ahmed

Language Proficiency

TOEFL: Overall Score: 110/120 (Reading: 30, Listening: 30, Speaking: 23, Writing: 27)

Standardized Test Scores

GRE: Total: 311 (Quantitative: 164, Verbal: 147)

Research Interests

- Machine Learning
- Density Functional Theory
- Energy Conversion
- Molecular Dynamics
- Energy Storage Devices
- Electrochemistry

Publications

[1] Abdul Hamid Rumman, Kaushik Barua, **Shohaib Ibne Monju**, Mohd Rakibul Hasan Abed, Sadika Jannath Tan-Ema, Jafar F. Al-Sharab, Saquib Ahmed;

“Classification of CoCr-based magnetic thin films via GLCM texture features extracted from EFTEM images and machine learning,” *AIP Advances*, vol. 14, no. 11, p. 115017, Nov. 2024. [DOI](#)

[2] Miah Abdullah Sahriar, Abdul Hamid Rumman, Ahammad Ullah, Kaushik Barua, **Shohaib Ibne Monju**, Munim Shahriar Jawad, Md. Atik Faisal, Ridwan Radit Ahsan, Houk Jang, Saquib Ahmed;

“Optical image analysis of WSe₂ thresholding for layer detection,” *Computational Materials Science*, vol. 253, p. 113888, May 2025. [DOI](#)

Current Research Projects

Accelerated Design of High-Entropy Alloys via Probabilistic Substitution and Machine Learning with DFT Simulations

Collaborators: Abdul Hamid Rumman, Kaushik Barua, **Shohaib Ibne Monju**, Aabrar Bin Salahuddin, Md Tohidul Islam, Saquib Ahmed

– Developed machine learning classifiers (ANN, XGBoost) using elemental descriptors to predict HEA phase stability with high accuracy.

– Introduced a probabilistic substitution strategy to generate over 50,000 novel HEA compositions.

– Performed DFT simulations on selected alloys, confirming thermodynamic stability of most predicted structures.

Contribution: Writing – Original Draft, Methodology, Validation, Software, Formal Analysis

Progress: Manuscript under review at *Journal of Alloys and Compounds*

Technical Skills

- **Experimental Skills:**
Uniaxial Pressing, Sand Molding, Metallography, Microstructure Analysis, Chemical Analysis, Electrodeposition, Corrosion Testing.
- **Software & Computational Skills:**
VASP (DFT calculations), Origin Lab, CES Edupack (basic), MATLAB, SolidWorks, MS Office, LATEX.
Molecular Dynamics Simulation: – Basic Simulation using LAMMPS
– Structure Generation and Visualization: VESTA, OVITO, AtomsK, and Packmol.
- **Programming Languages:**
C, C++, Python (matplotlib, pandas, scikit-learn).

Extra-Curricular Activities

- Vice President, BUET Career Club (2023-2024)
- Publication Secretary, SAMME (Students' Association of Materials and Metallurgical Engineering)
- Part-time script evaluator at Udvash Academic & Admission Care
- Instructor at ScholaraiD Academic & Admission Aid
- Volunteer, Bangladesh Flood Disaster Relief Program

Honors and Awards

- Winner, 2020 ISCEA PTAK Prize (70% scholarship for ISCEA's Global Certification)
- Semifinalist, HULT Prize at BUET 2021

Scholarships

- Technical Scholarship (2020-2024), BUET
- Government Board Scholarships for SSC & JSC